

Younous Maillefer

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Education	Master in Mathematics EPFL, Lausanne, Switzerland	Sept 2023 – Feb 2026
	Bachelor in Physics EPFL, Lausanne, Switzerland	Sept 2019 – July 2023
Master Thesis	Bounds and structure of the cyclotomic shortness of Φ_1^N Grade: 6.0/6.0 Supervisor: Prof. Maryna Viazovska • Expert: Prof. James McKee - Investigated the cyclotomic shortness of $(z - 1)^N$. - Derived a strict exponential upper bound by using Euler's Pentagonal Number Theorem and Sudler's constant. - Applied the discrete Uncertainty Principle on finite abelian groups to identify necessary cyclotomic factors.	Autumn 2025
Semester Project	DFTK Refinement & Verification Grade: 5.5/6.0 Supervisor: Prof. Michael Herbst • Co-Supervisor: Bruno Ploumhans - Validated error estimates implementation in <code>DFTK.jl</code>. - Automated benchmarks on large datasets using the EPFL HPC cluster. - Analyzed trade-offs between kinetic energy cutoff and numerical accuracy.	Spring 2025
Mathematics	Graduate (Theoretical): Riemann Surfaces, Topology, Representation Theory, Graph Theory, Abstract Analysis on Groups, Introduction to Number Theory Graduate (Applied): Lattice Models, Error Control in Scientific Modeling, Optimization on Manifolds, Number Theory in Cryptography, Probability Methods in Modern AI Undergraduate: Commutative Algebra, Lie Algebra, Functional Analysis, Differential Geometry, Homology, Metric Spaces	
Physics	Undergraduate: Classical Electrodynamics, Quantum Physics, Analytical Mechanics, Solid-State Physics, Particle Physics, Computational Physics, Statistical Physics, Philosophy of Exact Sciences	
Computational Skills	Language: Julia (Advanced, <code>DFTK.jl</code>), Python (NumPy, PyTorch), MATLAB, C++, SageMath, Lean (currently following a PHD course at EPFL). Other: Machine Learning, Introduction to Neural Networks, HPC (Linux/Bash clusters), Git, $\LaTeX$.	
Teaching	Teaching Assistant (EPFL): Courses: Analysis I & II, Linear Algebra I & II, Classical Mechanics.	2021 – Present
	Substitute Teacher: Various Gymnasiums (Biel, Payerne, Neuchâtel). Taught mathematics and physics; prepared theory, exercises, and exams.	2018 – Present
Research	Master Valorisation (EPFL): Researching cyclotomic shortness for upcoming publication and studying artificial life.	Mars 2026 – Present
Languages	French: Native	German: B2+ English: C1